

Mycelium-Inspired Mutualistic Adaptive Learning Systems (MMALS): A Research Chronicle from Bio-Inspired Functional Memory to Auditable Inferred-Context Continual Learning

Guillaume Harbonnier

Draft research article and reproducibility package

Compiled May 2026

Abstract

This paper consolidates the research trajectory of Mycelium-Inspired Mutualistic Adaptive Learning Systems (MMALS), a bio-inspired continual-learning program that treats memory not as a static parameter snapshot but as a mediated functional exchange across specialists. The project began as a mycelium metaphor for routing and mutualistic exchange, then progressively became an experimental continual-learning harness with explicit baselines, functional drift diagnostics, inferred context, no-leakage policy selection, and audit artifacts. The central thesis that emerged is that the relevant memory object is not merely the selected route, but the function computed through that route: routes, latent transformations, outputs, context posteriors, and readout policies must be observed together. We review the sequence from v0.1 to v1.1-RC2H, including route-function disentanglement, comparative benchmarks, Hopfield and co-evolution controls, geometry and fluid-inspired diagnostics, inferred-context stress, and the recent RC2 selector lineage. Evidence to date supports the value of functional-memory diagnostics and guarded context-sensitive deployment policies, but does not yet justify production-readiness claims. The latest evidence identifies a stable deployable anchor family, while showing that free validation-based policy selection can still underperform. The next milestone is multi-dataset robust qualification across MNIST, FashionMNIST, RotatedMNIST, and PermutedMNIST, followed by v0.11-style comparative benchmarking with accuracy, forgetting, cost, memory, parameters, runtime, and auditability.

Claim status

This is a research article and packaging draft, not a peer-reviewed paper and not a production certification. The strongest current claim is: *MMALS has a promising, auditable inferred-context continual-learning mechanism with replay-beating evidence on FashionMNIST overlap-chain tests, but the automatic selector is not yet sufficiently reliable to replace safe anchored policies.*

Contents

1	Executive summary	4
2	Scientific motivation	4
3	Core architecture	5
3.1	Hosts, transformations, routes, and mediated latent state	5
3.2	Route drift, latent drift, and output drift	5

4	Benchmark setting	6
5	Research chronicle: from v0.1 to v1.1-RC2H	6
5.1	v0.1: structural routing toy model	6
5.2	v0.2: fairer baselines and capacity checks	7
5.3	v0.3: making the fungal medium real	7
5.4	v0.4: warm-up, delayed mutualism, and soft consolidation	7
5.5	v0.5: route memory and output distillation	7
5.6	v0.6: route memory under harder benchmarks	7
5.7	v0.7: functional route memory	8
5.8	v0.8: final evidence harness	8
5.9	v0.9: route-function disentanglement	8
5.10	v0.10: cost and scalability	9
5.11	v0.11: comparative evidence harness	9
5.12	v0.12: Hopfield and co-evolution controls	9
5.13	v0.13 and v0.13.1: co-evolution and warm weak CE	10
5.14	v0.14 family: inferred context transition	10
5.15	v1.0-RC1b: benchmark-ready inferred-context candidate	10
5.16	v1.1-RC2b: strict no-leakage policy selection	10
5.17	RC2c to RC2g: repairing the selector and boundary classes	11
5.18	RC2H: RC1b-anchored conservative selector	11
6	A unified interpretation	12
6.1	What is the memory object?	12
6.2	Why the selector became the bottleneck	13
7	Cost and engineering interpretation	13
8	Auditability and Hydro diagnostics	13
9	Failure analysis	14
9.1	RC2b failure: free strict selection over-selects fragile policies	14
9.2	RC2e failure: class 2 fixed, class 4 collapsed	14
9.3	RC2f failure: class 4 fixed, class 2 weak again	14
9.4	RC2g failure: tri-boundary balance improves smoke but not evidence	14
9.5	RC2H failure: conservative anchor helps but context-only overrides remain unsafe	14
10	Current best scientific statement	14
11	Next research roadmap	15
11.1	RC2I: dual-anchor conservative selector	15
11.2	Multi-dataset robust qualification	15
11.3	v0.11-style comparative campaign	15
12	Limitations	16
13	Conclusion	16
A	Milestone table with child-level story	17

B Reproducibility package contents	18
C Notation	18

1 Executive summary

The story of MMALS is the story of a metaphor becoming a measurement program. The early metaphor was biological: a fungal medium connects several hosts, routes energy, exchanges nutrients, and preserves useful adaptations. The machine-learning translation became a continual-learning system with several specialist hosts, transformations, a routing matrix, a mediated latent state, and memory losses that preserve old behavior while learning new tasks.

The research program evolved through four major scientific transitions.

- T1. From route memory to functional memory.** Early versions asked whether preserving the route was enough. The answer became no: the same route can compute a different function if hosts or transformations drift.
- T2. From explicit task IDs to inferred context.** Early versions knew the task. The v1 direction asks whether the model can infer the situation from the input and memory.
- T3. From performance-only notebooks to evidence harnesses.** v0.11 and later introduced baselines, cost, parameters, runtime, forgetting, and reproducibility artifacts.
- T4. From free selection to conservative deployment.** RC2b-RC2H showed that selecting a policy only by validation proxy can choose fragile policies. The next selector must be anchored on safe guarded policies and allow overrides only with strong evidence.

Story for a 12-year-old

Imagine a forest where different trees are good at different things. A mushroom network connects them. At first we asked: “Did the mushroom use the same path as before?” Later we learned that this is not enough. The path can be the same, but the message carried along the path can change. So the real question became: “Did the forest still solve the old problem in the same useful way while learning a new problem?”

2 Scientific motivation

Continual learning studies how a model can learn a sequence of tasks without catastrophically forgetting earlier tasks. Classical approaches include regularization methods such as Elastic Weight Consolidation (Kirkpatrick et al., 2017), distillation-based methods such as Learning without Forgetting (Li and Hoiem, 2017), replay methods, progressive neural networks (Rusu et al., 2016), exemplar-based incremental classifiers (Rebuffi et al., 2017), and gradient memory methods (Lopez-Paz and Ranzato, 2017; Chaudhry et al., 2019). These methods are useful comparators, but they often frame memory as parameters, examples, regularization, or architectural isolation.

MMALS explores a different analogy: biological mycelial networks. Mycorrhizal networks can connect organisms and support exchange of resources or signals (Simard et al., 1997); fungal growth and transport networks have also inspired network optimization models (Bebber et al., 2007). The analogy is not used as biological proof. It is used as an engineering prior: useful memory may emerge from mediated exchange across specialists rather than from one monolithic representation.

The architectural question is therefore:

Can a system of specialists, connected by a learned fungal medium, preserve useful functions while adapting to changing contexts?

The scientific question is stronger:

Can we measure, audit, and falsify that preservation in a no-leakage continual-learning benchmark?

Story for a 12-year-old

Suppose you have five friends. One is good at shoes, one at shirts, one at numbers, one at maps, and one at drawings. A mushroom messenger decides whose advice to combine. If the group learns a new game, we do not only check whether the messenger visited the same friend. We check whether the group still gives the right advice.

3 Core architecture

3.1 Hosts, transformations, routes, and mediated latent state

Let x be an input. MMALS uses N specialist hosts. Each host i produces a representation

$$g_i(x). \quad (1)$$

A fungal transformation T_i maps the host representation into a shared exchange space:

$$\tilde{z}_i = T_i(g_i(x)). \quad (2)$$

A learned router produces weights

$$r_i(x, c) = \frac{\exp(a_i(x, c)/\tau_r)}{\sum_j \exp(a_j(x, c)/\tau_r)}, \quad (3)$$

where c is a context signal. In early v0.x versions, c was effectively an explicit task identifier. In the v1 direction, c is inferred from x and memory. The mediated latent state is

$$z_f(x, c) = \sum_i r_i(x, c) T_i(g_i(x)). \quad (4)$$

A readout then predicts the class label from z_f .

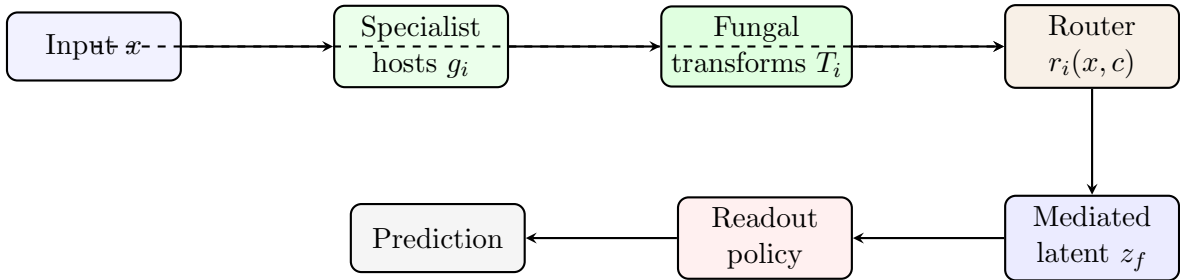


Figure 1: Conceptual MMALSflow. The route is only part of memory; the mediated latent function and readout behavior also matter.

3.2 Route drift, latent drift, and output drift

The v0.9 route-function disentanglement introduced three distinct measurements:

$$D_r = \mathbb{E}_{x,c} [\|r_{\text{current}}(x, c) - r_{\text{teacher}}(x, c)\|_2^2], \quad (5)$$

$$D_z = \mathbb{E}_{x,c} [\|z_{f,\text{current}}(x, c) - z_{f,\text{teacher}}(x, c)\|_2^2], \quad (6)$$

$$D_y = \mathbb{E}_{x,c} [D_{\text{KL}}(p_{\text{teacher}}(y|x, c) \| p_{\text{current}}(y|x, c))]. \quad (7)$$

A central non-implication is

$$r_{\text{current}}(x, c) \approx r_{\text{teacher}}(x, c) \not\Rightarrow z_{f,\text{current}}(x, c) \approx z_{f,\text{teacher}}(x, c). \quad (8)$$

Because the route can stay stable while the host or transformation changes, preserving route topology alone is insufficient.

Story for a 12-year-old

A bus can take the same road every day, but the passengers can be different. If the passengers are different, the trip has a different meaning. In MMALS, the route is the road, but the useful memory is the passengers plus the destination: what computation actually arrives.

4 Benchmark setting

The current main stress benchmark is an overlap-chain class-incremental setting:

$$(0, 1) \rightarrow (1, 2) \rightarrow (2, 3) \rightarrow (3, 4) \rightarrow (4, 5). \quad (9)$$

Labels are global class IDs. This makes the problem harder than disjoint binary task-incremental learning because classes such as 2 and 4 sit at boundaries between contexts. Later notebooks support MNIST, FashionMNIST, RotatedMNIST, and PermutedMNIST.

Runs are staged:

- **smoke:** one seed, small samples, execution and artifact sanity.
- **evidence:** three seeds, larger samples, intermediate scientific signal.
- **robust:** five seeds, same larger samples, stronger qualification.
- **paper:** larger samples and seeds, intended only after the candidate is frozen.

Story for a 12-year-old

Instead of five separate school quizzes, the teacher makes a chain: quiz 1 has animals 0 and 1, quiz 2 has 1 and 2, quiz 3 has 2 and 3. Now class 2 appears in different neighborhoods. The model must understand the situation, not just memorize the quiz number.

5 Research chronicle: from v0.1 to v1.1-RC2H

This section gives the full story in two layers: the PhD-level scientific role and the corresponding explanation for a 12-year-old.

5.1 v0.1: structural routing toy model

The first prototype established the basic idea: multiple hosts can be combined by a learned router, and the router can be interpreted as a mycelium-like medium. This was not yet evidence of continual learning. It was an architectural sketch.

Story for a 12-year-old

We built the first tiny forest. There were several trees and a mushroom path connecting them. The first question was simple: can the mushroom choose which tree to listen to?

5.2 v0.2: fairer baselines and capacity checks

The second step asked whether the apparent benefit was just capacity. The model needed to be compared against simpler neural baselines with comparable or at least explicit parameter counts. This began the discipline that later became central: never compare a mycelium-inspired model only against weak strawmen.

Story for a 12-year-old

If one team wins a race with a giant engine and the other team has a bicycle, the race is unfair. v0.2 started checking whether MMALS was winning because of the idea, or just because it had more parts.

5.3 v0.3: making the fungal medium real

The fungal medium became more than a metaphor. The architecture began to include specialists, transformations, routes, a mediated latent state, and mutualistic signals. The question became whether a learned medium can stabilize exchanges across hosts.

Story for a 12-year-old

The mushroom network stopped being just a drawing. It became a real messenger that could mix advice from different trees.

5.4 v0.4: warm-up, delayed mutualism, and soft consolidation

Early training can be fragile. v0.4 introduced warm-up and delayed mutualism so that the system would not over-constrain itself before useful host behavior emerged. The lesson was that mutualism must be timed.

Story for a 12-year-old

You should not ask a new team to make perfect teamwork on day one. First everyone learns their job; then the mushroom starts asking them to cooperate.

5.5 v0.5: route memory and output distillation

The next hypothesis was that if the router remembers old routing behavior and the outputs remain close to a teacher, old tasks should be preserved. Output distillation gave a behavioral anchor; route memory gave a mechanistic anchor.

Story for a 12-year-old

The mushroom tried to remember two things: which path it used before, and what answer the forest gave before. This is like remembering both the road and the final answer.

5.6 v0.6: route memory under harder benchmarks

v0.6 asked a causal question: if output distillation is already strong, does route memory add value on harder data? This was the start of a more skeptical style. A mechanism is not accepted because it sounds biologically plausible; it must improve or explain results.

Story for a 12-year-old

If remembering the answer already works, does remembering the path help? v0.6 checked whether the path memory was useful or just decorative.

5.7 v0.7: functional route memory

v0.7 moved from route preservation toward preserving the computation through routes. This is where the project began to converge on mediated functional memory: the system should preserve the old latent function, not merely the same route probabilities.

Story for a 12-year-old

We learned that it is not enough to say, “I used the same road.” We must ask, “Did the road still deliver the same package?”

5.8 v0.8: final evidence harness

v0.8 consolidated earlier variants into an evidence harness. It asked which variant performed best and exported results in a more reproducible form. This was the first move from exploratory notebooks to evidence packaging.

Story for a 12-year-old

Instead of trying one mushroom trick at a time, v0.8 organized a tournament and wrote down the scores.

5.9 v0.9: route-function disentanglement

v0.9 is a major conceptual step. It explicitly distinguishes route drift, functional latent drift, and output drift. It shows that route-only memory can fail badly even when more functional variants remain stable. In the v0.9 scorecard, the route-only control reached about 0.696 final accuracy with about 0.295 forgetting, while full functional memory reached about 0.991 final accuracy and near-zero forgetting.

The conclusion is not that route information is useless. It is that route information alone is not the memory object. Functional memory must be defined at the level of mediated latent computation and behavior.

Table 1: v0.9 route-function diagnostic summary.

Variant	Final accuracy	Forgetting
Full functional memory	0.9908	0.0004
Distillation only	0.9913	0.0004
Latent only	0.9905	0.0010
Route plus latent	0.9903	0.0011
Route memory only	0.6959	0.2951

Story for a 12-year-old

The mushroom remembered the map, but the forest still gave wrong answers. That proved the map alone is not memory. The answer depends on what travels through the map.

5.10 v0.10: cost and scalability

v0.10 made the cost question explicit. The model was larger and slower than simple MLP baselines. The scientific trade-off became clear: MMALS must justify its complexity through robustness, auditability, context handling, and functional memory, not just raw accuracy.

Story for a 12-year-old

A big forest is expensive to maintain. If it is only a little better than a small garden, maybe it is not worth it. v0.10 asked: what do we get for the extra trees?

5.11 v0.11: comparative evidence harness

v0.11 compared MMALS against known continual-learning baselines. In a smoke configuration, full functional memory reached 0.967 final accuracy and 0.003 forgetting, behind joint upper bound, PNN, and experience replay, but ahead of LwF, EWC, fine-tuning, and sparse MoE in that run. It used about 1.23M parameters versus about 218k for the MLP baselines.

Table 2: v0.11 comparative smoke result, selected rows.

Method	Final acc.	Forgetting	Params	Train time (s)
Joint upper bound	0.980	0.000	218k	1.02
PNN	0.979	0.000	218k	0.37
Experience replay	0.971	0.002	218k	0.49
MMALSfull functional memory	0.967	0.003	1.23M	3.99
LwF	0.962	0.003	218k	0.65
EWC	0.873	0.102	218k	0.46
Fine-tuning	0.821	0.142	218k	0.41
Sparse MoE	0.795	0.176	1.19M	1.72

The key lesson was honest: MMALS was not automatically the best on simple benchmarks. Its value proposition must be harder settings, inferred context, auditability, and functional memory.

Story for a 12-year-old

The mushroom forest did well, but it did not beat every simpler team. That is okay. It told us the mushroom idea must prove itself on harder races, not just easy ones.

5.12 v0.12: Hopfield and co-evolution controls

The Hopfield branch tested whether memory should be understood as a frozen landscape. The evidence suggested that hard frozen-landscape memory is not the main answer. Hopfield-style ideas remain useful as diagnostic controls, but the deployable path remains functional mediated adaptation rather than simply freezing transformations.

Story for a 12-year-old

We tried turning part of the forest into stone so it would never change. It remembered some things, but it was not the best way to keep learning. A living forest needs some stable paths and some flexible growth.

5.13 v0.13 and v0.13.1: co-evolution and warm weak CE

The co-evolution branch explored transform drift, Jacobian drift, and resistance-like quantities inspired by geometry and physical systems. Warm weak CE ablations showed diagnostic value but did not yet establish a superior training recipe. The research lesson was that geometry can be a microscope before it becomes an optimizer.

Story for a 12-year-old

We gave the mushroom a microscope to watch how the paths bend and stretch. The microscope helped us see problems, but it was not yet the best tool to fix them.

5.14 v0.14 family: inferred context transition

The major conceptual shift before v1.0 was context inference. Earlier versions were told the task. The v1 direction asks the model to infer the situation from the input and memory, for example through a posterior $q(c|x)$. A small group of examples can be compared with old memory prototypes to infer context.

Story for a 12-year-old

Before, someone told the mushroom, “This is math class.” Now the mushroom must look around: rulers, triangles, numbers on the board. It has to guess: “This is probably math class.”

5.15 v1.0-RC1b: benchmark-ready inferred-context candidate

v1.0-RC1b consolidated a proto-first MMALSSbackbone, validation-selected context temperature/blend, and a guarded global head. It made the jump from explicit task IDs toward inferred-context class-incremental stress. The best candidate family became the guarded context-gap policy, often represented as `paired_context_gap_selected` or `context_temp_blend_selected_head_guard_035`.

Story for a 12-year-old

The mushroom no longer gets a label saying which classroom it entered. It looks at the objects, compares them with old memories, and chooses the path that seems most likely.

5.16 v1.1-RC2b: strict no-leakage policy selection

RC2b introduced a strict policy-selection protocol. It trains the same backbone, splits memory into repair, context-selection, and policy-validation partitions, builds candidate policies, scores them on validation memory only, selects before final-test evaluation, and exports audit traces. This is a major methodological improvement.

In the RC2b FashionMNIST robust run, the fixed RC1b context-gap candidate reached 0.9010 final average accuracy, 0.8183 min-task accuracy, 0.0556 forgetting, and 0.8235 class-2 accuracy. The RC2b automatic selected policy reached 0.8880 final average accuracy and still beat replay, but failed to match fixed RC1b.

Table 3: RC2b robust FashionMNIST selected results.

Method	Final acc.	Min task	Forgetting	Class 2	Class 5
Fixed RC1b context-gap	0.9010	0.8183	0.0556	0.8235	0.9985
RC2b selected policy	0.8880	0.7773	0.0644	0.7848	0.9985
Experience replay	0.8390	0.6985	0.1352	0.6188	0.9970

Story for a 12-year-old

The mushroom made a rule: before looking at the final exam, it chooses its strategy using only practice memories. That is fair. But sometimes it chose the wrong strategy, so the rule was fair but not yet smart enough.

5.17 RC2c to RC2g: repairing the selector and boundary classes

RC2c through RC2g were debugging branches:

- RC2c added risk-aware penalties, class-2 and min-task floors, no-repair vetoes, and hysteresis.
- RC2d aligned the selector to gates and fixed Drive archiving.
- RC2e made the selector NaN-safe and improved class-2 behavior.
- RC2f protected class 4 after class-4 collapse.
- RC2g introduced tri-boundary equilibrium across classes 2, 4, and 5.

RC2g smoke looked promising, but RC2g evidence showed the danger of free validation selection. The fixed RC1b context-gap candidate reached about 0.9058 final accuracy in evidence mode, while RC2g selected about 0.8704. The selector, not the backbone, was the issue.

Story for a 12-year-old

The mushroom kept trying new voting rules. One rule protected class 2 but hurt class 4. Another protected class 4 but hurt class 2. RC2g tried to balance the three difficult children: 2, 4, and 5. It helped, but the mushroom still sometimes picked a worse rule than the old safe one.

5.18 RC2H: RC1b-anchored conservative selector

RC2H inverted the selection question. Instead of asking which candidate has the highest proxy score, it asks whether there is enough validation-memory evidence to override the known-good RC1b anchor. It uses the guarded context-gap family as a default deployable anchor and permits overrides only when safety gates are satisfied.

In FashionMNIST evidence mode, RC2H validated the infrastructure but did not yet succeed as a performance selector. The true global guard was best in that run, the RC1b context-gap anchor was second, and RC2H selected policy remained replay-beating but weaker than the best deployable candidate.

Table 4: RC2H evidence FashionMNIST summary.

Method	Final acc.	Min task	Forgetting	Class 2	Class 5
True global guard	0.8957	0.8058	0.0623	0.8863	0.9958
RC1b context-gap anchor	0.8858	0.7908	0.0772	0.8429	0.9958
RC2H selected policy	0.8788	0.7358	0.0958	0.7696	0.9942
Experience replay	0.8361	0.6838	0.1415	0.6054	0.9975

The lesson is that the anchor should likely become a dual safe-anchor family: context-gap plus true global guard. Context-only global candidates should remain audit candidates, not deployment overrides, unless much stronger evidence exists.

Story for a 12-year-old

The mushroom learned a safer rule: “Do not change from the trusted plan unless the new plan is clearly better.” But two times it still trusted a shortcut that looked good in practice and was weaker on the final test. Next, the mushroom should trust two safe teachers instead of one.

6 A unified interpretation

6.1 What is the memory object?

The main scientific result of the research program is not a single accuracy number. It is the clarification of the memory object. MMALSmemory has at least five layers:

1. **Route memory:** which hosts are weighted.
2. **Transformation memory:** how host signals are mapped into exchange space.
3. **Latent functional memory:** what mediated representation z_f is computed.

4. **Output behavioral memory:** whether predictions remain stable.
5. **Context-posterior memory:** whether the system infers the right situation.

A deployment policy adds a sixth layer: the readout/head and its calibration. The RC2 lineage showed that even a strong backbone can be undermined by a weak selector or readout policy.

Story for a 12-year-old

Memory is not just “which road did I take?” It is the road, the package, the messenger, the final answer, and whether the mushroom understood which village it was visiting.

6.2 Why the selector became the bottleneck

RC2b through RC2H show a repeated pattern. The backbone is often capable; diagnostic probes reach high accuracy; fixed guarded policies perform well. But validation-memory policy selection can overfit proxy signals. A policy can look good on validation memory but damage weak boundary classes in final evaluation.

This creates a deployment principle:

No-leakage selection is necessary but not sufficient. A selector must also be conservative, anchor-aware, and weak-class aware.

Story for a 12-year-old

Not cheating on the exam is necessary. But even if you do not cheat, you can still choose the wrong study strategy. RC2H is about choosing a strategy carefully, not just fairly.

7 Cost and engineering interpretation

MMALS is not free. The v0.11 and RC2 runs make parameter cost visible: MLP baselines are around 218k parameters, while MMALS variants are around 1.2M to 1.34M parameters. The deployment question is therefore not “does MMALS beat everything on every toy benchmark?” It is:

Does MMALS provide enough robustness, context inference, auditability, and functional-memory control to justify its higher cost in harder continual-learning settings?

Table 5: Cost-oriented comparison from selected evidence packages.

Method family	Example final acc.	Params	Model size
MLP replay baseline	0.836–0.971	~218k	~0.83 MB
PNN baseline	0.979 in v0.11 smoke	~218k	~0.83 MB
Sparse MoE	0.795 in v0.11 smoke	~1.19M	~4.54 MB
MMALSfunctional memory	0.967 in v0.11 smoke	~1.23M	~4.70 MB
MMALSRC2 family	0.879–0.901 in harder overlap-chain	~1.34M	~5.10 MB

Story for a 12-year-old

A big forest costs more than a small garden. So the big forest must do something the garden cannot: explain itself, survive harder weather, and remember old paths while growing new ones.

8 Auditability and Hydro diagnostics

The v1.1 Hydro branch adds a fluid-inspired diagnostic layer: route drift, latent drift, output drift, adaptive viscosity, Reynolds-like instability, turbulence-like scores, and wave/context entropy. The current conclusion is conservative. Hydro is useful as an audit layer, not yet as the default performance policy.

Hydro metrics matter because they create observability. A model that learns continually should not only output an accuracy number; it should explain whether routes, latent states, outputs, and context posterior were stable or turbulent.

Story for a 12-year-old

A river can look calm or turbulent. Hydro gives the mushroom forest a water-meter. It tells us whether the learning flow is smooth or chaotic. For now it is a dashboard, not the steering wheel.

9 Failure analysis

The recent selector lineage produced valuable failures.

9.1 RC2b failure: free strict selection over-selects fragile policies

RC2b was methodologically correct but not performance-optimal. It selected one policy per checkpoint without final-test leakage, but selected weaker policies too often.

9.2 RC2e failure: class 2 fixed, class 4 collapsed

Class-specific rescue can overcorrect. Protecting one boundary class can sacrifice another.

9.3 RC2f failure: class 4 fixed, class 2 weak again

Boundary protection must be symmetric and equilibrium-oriented.

9.4 RC2g failure: tri-boundary balance improves smoke but not evidence

A selector can pass smoke and still fail evidence. Smoke is execution validation, not scientific validation.

9.5 RC2H failure: conservative anchor helps but context-only overrides remain unsafe

Anchoring is necessary, but the safe anchor set should include both context-gap and true global guard. Context-only global candidates should be restricted.

Story for a 12-year-old

Every failed version taught us which child fell down: class 2, class 4, class 5, or the selector itself. The failures were not useless. They were like footprints showing where the bridge is weak.

10 Current best scientific statement

A careful statement after RC2H is:

MMALShas demonstrated a replay-beating, auditable inferred-context continual-learning framework on FashionMNIST overlap-chain experiments. Fixed guarded policies, especially the RC1b context-gap family and the true CE/KL guarded global head, remain stronger than the current automatic selectors. The next deployable selector should be dual-anchor conservative, allowing overrides only from guarded policies with strong validation-memory evidence and weak-class safety gates. Multi-dataset robust qualification remains necessary before stronger claims.

Story for a 12-year-old

The mushroom forest is promising. It beats some simple memory tricks, and it keeps a good notebook of what happened. But it is not ready to drive alone. It still needs a safe rule: trust the strongest known teachers unless a new plan is clearly safer and better.

11 Next research roadmap

11.1 RC2I: dual-anchor conservative selector

The next notebook should implement a dual-anchor rule:

- safe anchors: `context_gap_selected`, `context_temp_blend_selected_head_guard_035`, and `proto_global_head`
- context-only global candidates remain in audit but cannot override by default in evidence/robust;
- overrides must be guarded, margin-positive, class-safe, min-task-safe, and forgetting-safe;
- report rejection reasons explicitly.

11.2 Multi-dataset robust qualification

After RC2I passes FashionMNIST evidence and robust, it should be evaluated on:

1. MNIST overlap-chain;
2. FashionMNIST overlap-chain;
3. RotatedMNIST overlap-chain;
4. PermutedMNIST overlap-chain.

11.3 v0.11-style comparative campaign

The final campaign must include not only accuracy and forgetting but also:

- parameters and model size;
- training time and memory;
- replay buffer items;
- min-task and weak-class floors;
- context posterior metrics;
- Hydro audit metrics;
- no-leakage selection traces;
- seed-level weak-run diagnosis.

Story for a 12-year-old

The next big tournament should not only ask who wins. It should ask who wins, who forgets, who is expensive, who explains their decisions, and who stays calm when the game changes.

12 Limitations

This research program has important limitations:

1. Most claims are still MNIST-family claims.
2. Smoke runs are not evidence.
3. Evidence runs are not robust qualification.
4. The selector remains the main bottleneck.
5. The model is more expensive than MLP baselines.
6. Biological inspiration is an analogy, not a biological validation.
7. No claim of production readiness, safety certification, open-world learning, or industry level qualification is justified yet.

Story for a 12-year-old

The mushroom forest has not traveled the whole world yet. It has mostly practiced on small maps. Before we trust it in a real jungle, it needs harder maps, more races, and stricter judges.

13 Conclusion

MMALS began as a mycelium-inspired routing architecture and matured into an auditable continual-learning research program. The strongest conceptual contribution is the distinction between route memory and functional memory. The strongest engineering contribution is the evidence harness: no-leakage policy selection, replay comparisons, weak-class diagnostics, context posterior auditing, Hydro stability traces, Drive persistence, manifests, and reproducible packages.

The current state is promising but unfinished. The model has credible replay-beating behavior in the harder overlap-chain setting, but automatic policy selection is not yet reliable enough to be the default deployable mechanism. The next step is not another unconstrained selector. It is a dual-anchor conservative selector followed by multi-dataset robust qualification and a v0.11-style comparative cost/performance campaign.

Story for a 12-year-old

The mushroom forest has learned a lot. It can remember, compare, and explain more than before. But the wisest mushroom is not the one that always changes paths. It is the one that knows when to keep the safe path and when a new path is truly better.

A Milestone table with child-level story

Table 6: MMALS research progression.

Step	Scientific role	Story for a 12-year-old
v0.1	Structural routing toy model.	Build the first mushroom path between smart trees.
v0.2	Fair baselines and capacity checks.	Make sure the mushroom is not winning just because it has a bigger engine.
v0.3	Real fungal medium with hosts and transformations.	The mushroom messenger starts mixing advice from different trees.
v0.4	Warm-up and delayed mutualism.	Let each tree learn its job before forcing teamwork.
v0.5	Route memory and output distillation.	Remember both the road and the answer.
v0.6	Route-memory ablation on harder data.	Check whether remembering the road really helps.
v0.7	Functional route memory.	Check whether the package delivered by the road stayed the same.
v0.8	Evidence harness.	Organize a tournament and record the scores.
v0.9	Route-function disentanglement.	The same road can carry a different package.
v0.10	Cost and scalability.	A big forest must justify its cost.
v0.11	Comparative baselines.	Race the mushroom against known teams.

Step	Scientific role	Story for a 12-year-old
v0.12	Hopfield/frozen controls.	Test whether turning the forest to stone helps.
v0.13	Geometry and co-evolution diagnostics.	Use a microscope to see how paths stretch.
v0.14	Inferred context.	Guess the classroom without being told the subject.
v1.0-RC1b	Proto-first inferred-context candidate with guarded readout.	Choose paths by comparing today with old memories.
RC2b	Strict no-leakage policy selection.	Pick a strategy before looking at the final exam.
RC2c-RC2g	Risk-aware and boundary-safe selector repairs.	Stop one child from falling without pushing another down.
RC2H	RC1b-anchored conservative selector.	Keep the trusted plan unless a new plan clearly wins.
RC2I next	Dual-anchor conservative selector.	Trust two safe teachers before accepting a new shortcut.

B Reproducibility package contents

The accompanying GitHub-ready package contains:

- `paper/main.tex`: LaTeX source for this article;
- `paper/references.bib`: references;
- `paper/MMALS_Full_Story_PhD_Article.pdf`: compiled PDF;
- `metrics/`: compact CSV summaries extracted from v0.9, v0.11, RC2b, RC2G, and RC2H evidence packages;
- `notebooks/`: latest RC2H notebook when available;
- `docs/`: selected evidence PDFs;
- `README.md`: GitHub publication instructions.

C Notation

Symbol	Meaning
x	input
c	context, explicit in v0.x and inferred in v1 direction
$g_i(x)$	host representation
T_i	fungus transformation for host i
$r_i(x, c)$	route weight for host i
$z_f(x, c)$	mediated latent state
D_r	route drift
D_z	functional latent drift
D_y	output drift
$q(c x)$	inferred context posterior

References

- Bebber, D. P., Hynes, J., Darrah, P. R., Boddy, L., and Fricker, M. D. (2007). Biological solutions to transport network design. *Proceedings of the Royal Society B*, 274, 2307–2315.
- Bronstein, M. M., Bruna, J., LeCun, Y., Szlam, A., and Vandergheynst, P. (2017). Geometric deep learning: going beyond Euclidean data. *IEEE Signal Processing Magazine*, 34(4), 18–42.
- Chaudhry, A., Ranzato, M., Rohrbach, M., and Elhoseiny, M. (2019). Efficient lifelong learning with A-GEM. *ICLR*.
- Hopfield, J. J. (1982). Neural networks and physical systems with emergent collective computational abilities. *PNAS*, 79(8), 2554–2558.
- Kirkpatrick, J. et al. (2017). Overcoming catastrophic forgetting in neural networks. *PNAS*.
- Li, Z. and Hoiem, D. (2017). Learning without forgetting. *IEEE TPAMI*.
- Lopez-Paz, D. and Ranzato, M. (2017). Gradient episodic memory for continual learning. *NeurIPS*.
- Rebuffi, S.-A., Kolesnikov, A., Sperl, G., and Lampert, C. H. (2017). iCaRL: Incremental classifier and representation learning. *CVPR*.
- Rusu, A. A. et al. (2016). Progressive Neural Networks. *arXiv:1606.04671*.
- Simard, S. W. et al. (1997). Net transfer of carbon between ectomycorrhizal tree species in the field. *Nature*, 388, 579–582.
- van de Ven, G. M., Tuytelaars, T., and Tolias, A. S. (2022). Three types of incremental learning. *Nature Machine Intelligence*.
- Harbonnier, G. (2026a). *MMALS Prototype 0.9: Route-Function Disentanglement and Functional Memory Validation*. Internal research notebook and evidence package.
- Harbonnier, G. (2026b). *MMALS Prototype 0.11: Comparative Evidence Harness*. Internal research notebook and evidence package.

Harbonnier, G. (2026c). *MMALS v1.1-RC2b: Strict No-Leakage Policy-Selection Harness*. Internal research notebook and evidence package.

Harbonnier, G. (2026d). *MMALS v1.1-RC2h: RC1b-Anchored Conservative Selector Harness*. Internal research notebook and evidence package.